

Problem 1:

Let x_0 and x_1 be points in a path-connected space X . In class, we showed that any path α from x_0 to x_1 defines an isomorphism $\hat{\alpha} : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$. Prove that $\pi_1(X, x_0)$ is abelian if and only if for every pair of paths α, β from x_0 to x_1 , the isomorphisms $\hat{\alpha}$ and $\hat{\beta}$ agree.

Proof.

Assume $\pi_1(X, x_0)$ is abelian, since $\pi_1(X, x_0) \cong \pi_1(X, x_1)$ this implies that $\pi_1(X, x_1)$ is also abelian.

Let $[f] \in \pi_1(X, x_0)$

Consider

$$\begin{aligned}\hat{\alpha}([f]) &= [\alpha^{-1} * f * \alpha] \\ &= [\alpha^{-1} * \beta * \beta^{-1} * f * \alpha] \\ &= [\alpha^{-1} * \beta] * [\beta^{-1} * f * \alpha]\end{aligned}$$

We can split them up like this since $\alpha^{-1} * \beta$, and $\beta^{-1} * f * \alpha$ are loops of x_1

$$\begin{aligned}&= [\beta^{-1} * f * \alpha] * [\alpha^{-1} * \beta] \\ &= [\beta^{-1} * f * \alpha * \alpha^{-1} * \beta] = [\beta^{-1} * f * \beta] = \hat{\beta}([f])\end{aligned}$$

Assume $\hat{\alpha} = \hat{\beta}$.

Let $[f], [g] \in \pi_1(X, x_0)$. Let $\beta = f * \alpha$ which is a path from x_0 to x_1 .

Consider

$$\begin{aligned}\hat{\alpha}([f * g]) &= \hat{\beta}([f * g]) \\ &= [\alpha^{-1} * f^{-1} * f * g * f * \alpha] \\ &= [\alpha^{-1} * g * f * \alpha] \\ &= \hat{\alpha}([g * f])\end{aligned}$$

This implies that $[f] * [g] = [g] * [f]$ since α is an isomorphism.

Thus, $\pi_1(X, x_0)$ is abelian ■

Problem 2:

Let X be a path-connected space. Prove that $\pi_1(X)$ is trivial if and only if every map $S^1 \rightarrow X$ extends to a map $D \rightarrow X$, where D is the closed unit disk (with boundary S^1).

Proof.

Assume $\pi_1(X)$ is trivial. Let $f : S^1 \rightarrow X$ be a loop at x_0 . Let f be in terms of polar cords, so $f(\theta) = x$ for some $x \in X$. Since $\pi_1(X)$ is trivial this implies there exists a homotopy from f to the constant map on x_0 . Call it $h : S^1 \times [0,1] \rightarrow X$ s.t. $h(\theta,0) = f(\theta)$ and $h(\theta,1) = x_0$.

Define a map $F : D \rightarrow X$ s.t. by $F(\theta, r) = h(\theta, 1 - r)$, and for $F|_{S^1} = F(\theta, 1) = f(\theta)$. Thus, F is an extension of f .

Assume every map $S^1 \rightarrow X$ extends to a map $D \rightarrow X$. Let $f : S^1 \rightarrow X$ be a loop based at x_0 . By assumption we have a continuous map $F : D \rightarrow X$ s.t. $F|_{S^1} = f$.

This is by definition a homotopy since $F(\theta, 1) = f(\theta)$ and $F(\theta, 0) = x$ is constant for some $x \in X$. Thus, this implies that f is homotopic to a constant map which implies that $\pi_1(X, x_0)$ is trivial, and since X is path connected this implies that $\pi_1(X, x)$ is trivial for all $x \in X$. Thus, $\pi_1(X)$ is trivial. ■

Problem 3:

Let $p : \tilde{X} \rightarrow X$ be a covering map, and assume that $\tilde{X}$ is path-connected. Show that the cardinality of the preimage $p^{-1}(x)$ of a point $x \in X$ does not depend on x .

Proof.

Since $p(\tilde{X}) = X$ this implies that X is also path connected. Since for $x, y \in X$ then consider $\tilde{x} \in p^{-1}(x)$ and $\tilde{y} \in p^{-1}(y)$. Then $\tilde{X}$ is path connected, so there is a continuous path f from $\tilde{x}$ to $\tilde{y}$, and thus $q \circ f$ is a continuous path from x to y in X .

Let $x_0, x_1 \in X$. Let f be a path from x_0 to x_1

Define: $\varphi : p^{-1}(x_0) \rightarrow p^{-1}(x_1)$ s.t. $\varphi(x) = f_x(1)$ where f_x is the unique path lift of f beginning at x .

Well-Defined:

Since $p(\varphi(x)) = p(f_x(1)) = f(1) = x_1 \implies \varphi(x) \in p^{-1}(x_1)$.

Consider the reverse path of f call it g from x_1 to x_0 where $g(t) = f(1 - t)$

Now define $\varphi^{-1} : p^{-1}(x_1) \rightarrow p^{-1}(x_0) = g_x(1)$ where g_x is the unique path lift of g beginning at x .

Let $x \in p^{-1}(x_0)$.

Consider $\varphi^{-1}(\varphi(x)) = g_{f_x(1)}(1)$.

Since $p \circ g_{f_x(1)} = g$, $p \circ f_x(1 - t) = f(1 - t) = g$, and $f_x(1) = g_{f_x(1)}(0)$ since lifting paths is unique this implies that $g_{f_x(1)} = f_x(1 - t)$.

Thus, $g_{f_x(1)}(1) = x$.

Now the same argument works for $\varphi(\varphi^{-1}(y)) = y$ for $y \in p^{-1}(x_1)$.

Thus, $|p^{-1}(x_0)| = |p^{-1}(x_1)|$ ■

Problem 4:

Let $p : \tilde{X} \rightarrow X$ be a covering map with countable degree (see the previous problem). Prove that $\tilde{X}$ is an n -manifold if and only if X is an n -manifold.

Claim

Let $p : \tilde{X} \rightarrow X$ be a covering map then p is an open map.

Proof.

Let U be an open set in $\tilde{X}$. Let $x \in p(U)$

Since p is a covering map this implies that there is an open neighborhood V of x that is evenly covered. Thus, there exists an open set in W s.t. $p|_W : W \rightarrow V$ is a homeomorphism.

Thus, consider $U \cap W$ which is open in $\tilde{X}$ thus $p|_W (U \cap W) = p(U \cap W) = p(U) \cap p(W) \subset p(U)$ is an open neighborhood of x in X . Thus, $p(U)$ is open. ■

Claim

If X is locally homeomorphic to $\mathbb{R}^n$ then X is T_1

Proof.

Let $x, y \in X$ s.t. $x \neq y$. Let U be a neighborhood of x that is homeomorphic to an open subset of $\mathbb{R}^n$.

If $y \notin U$ we are done.

If $y \in U$ then since U is Hausdorff since it is homeomorphic to a subspace of a Hausdorff space we now that we can find an open neighborhood $V \subset U$ of x s.t. $y \notin V$.

Since V is open in an open set it is open in X .

Now for all $x \in X \setminus \{y\}$ we can find open sets of X that do not contain y and contain x .

Thus, this implies that $X \setminus \{y\}$ is open. Thus, $\{y\}$ is closed. Thus, X is T_1 . ■

Claim

Let $p : \tilde{X} \rightarrow X$ be a covering map. If U is evenly covered by p then every open subset of U is evenly covered.

Proof.

Let U be evenly covered by p . Let $V \subset U$ be open. Consider $p^{-1}(U) = \bigsqcup_{\alpha \in J} V_\alpha$ s.t. $V_\alpha \cong U$ this implies that $V_\alpha \cap p^{-1}|_{V_\alpha}(V) \cong V$.

Thus, $p^{-1}(V) = \bigsqcup_{\alpha \in J} V_\alpha \cap p^{-1}|_{V_\alpha}(V)$. Thus, open subsets of evenly covered sets are evenly covered. ■

Proof.

Assume $\tilde{X}$ is an n -manifold.

Let $x \in X$. Since p is a covering map this implies that there exists a neighborhood U of x that is evenly covered by p .

Thus, there exists an open set V in $\tilde{X}$ s.t. $p|_V V \rightarrow U$ is a homeomorphism.

Let $\tilde{x} = p^{-1}|_V(x) \in V$ this implies that there exists an open neighborhood W of $\tilde{x}$ s.t. W is homeomorphic to an open set of $\mathbb{R}^n$. Thus, consider $W \cap V$ since this is an open in W this implies that it is homeomorphic to an open subset of $\mathbb{R}^n$. Therefore, $p|_V(V \cap W)$ is an open neighborhood of x that is homeomorphic to an open subset of $\mathbb{R}^n$.

Let $\tilde{\beta}$ be a countable basis for $\tilde{X}$. Consider $\beta = p(\tilde{\beta})$ this is a collection of open sets since p is an open map.

Let $x \in X$. Since p is surjective this implies there is a $z \in \tilde{X}$ s.t. $p(z) = x$. Since $\tilde{\beta}$ is a basis there is a $\tilde{B} \in \tilde{\beta}$ s.t. $z \in \tilde{B}$. Thus, $x \in p(\tilde{B}) = B \in \beta$.

Thus, β covers X .

Let $x \in B_1 \cap B_2$ for $B_1, B_2 \in \beta$. Since p is surjective there is a $z \in \tilde{X}$ s.t. $p(z) = x$. Since p is continuous this implies that $p^{-1}(B_1 \cap B_2)$ is an open neighborhood of z . Thus, there exists a basis set $\tilde{B}_3$ of z s.t. $z \in \tilde{B}_3 \subset p^{-1}(B_1 \cap B_2)$. Thus, $x \in p(\tilde{B}_3) = B_3 \subset B_1 \cap B_2$.

Thus, β is a countable basis for X , thus X is second countable.

Let $x, y \in X$ s.t. $x \neq y$. Proven above since X is locally homeomorphic to $\mathbb{R}^n$ we now that X is T_1 .

Thus, $p^{-1}(x)$ and $p^{-1}(y)$ are disjoint closed sets in $\tilde{X}$. Since $\tilde{X}$ is a n -manifold it is metrizable thus normal. Thus, we can find two disjoint open sets N_x and N_y s.t. $p^{-1}(x) \subset N_x$ and $p^{-1}(y) \subset N_y$.

Let U be an evenly covered neighborhood of x . Let $U_\alpha \subset p^{-1}(U)$ be one of the disjoint homeomorphic copies of U that make up $p^{-1}(U)$. Let $\{x_\alpha\} = U_\alpha \cap p^{-1}(x)$. WLOG let U_α be path connected since $\tilde{X}$ is locally path connected since it is an n -manifold, so we can just intersect U_α with a path connected neighborhood of x_α .

Consider $p(U_\alpha \cap N_x) \subset U$. Since $U_\alpha \cap N_x$ is open this implies that $p(U_\alpha \cap N_x)$ is an open subset of U . Thus, $p(U_\alpha \cap N_x) = A$ is evenly covered.

Therefore, $p^{-1}(A) = \bigsqcup_{\alpha \in J} V_\alpha$ s.t. $V_\alpha \cong U_\alpha \cap N_x$. Thus, V_α is also path connected.

Since $x_\alpha \in U_\alpha \cap N_x \implies x \in A \implies p^{-1}(x) \subset p^{-1}(A)$.

Also since $p^{-1}(y) \cap N_x = \emptyset \implies y \notin A$.

Now let V be an evenly covered neighborhood of x , and do the same thing as above with $V_\beta \subset p^{-1}(V)$ and call $B = p(V_\beta \cap N_y)$.

Assume for contradiction that $A \cap B \neq \emptyset$. Let $z \in A \cap B$. Let $p(z_a) = z$ for $z_a \in U_\alpha \cap N_x$. Since A is path connected let f be a path from $x \rightarrow z$ and same for B let g be a path from $z \rightarrow y$. Consider $f * g$

Thus, there exists a unique lift $\widetilde{f * g} = \tilde{h}$ that is based at $p^{-1}|_{U_\alpha \cap N_x}(x)$.

Since $p(\tilde{h}(t)) = f(2t) \in A$ for $t \in [0, 1/2]$ this implies that since p restricted to one of the covering sets of A is a homeomorphism that,

$$p|_{U_\alpha \cap N_x}(\tilde{h}(1/2)) = f(1) = z \implies \tilde{h}(1/2) = z_a.$$

Now do the same for $p(\tilde{h}(t)) = g(2t) \in B$ for $t \in [1/2, 1]$. This implies that there exists a homeomorphic copy of V_β called $U_\beta \subset p^{-1}(V_\beta)$ s.t.

$$\tilde{h}(1/2) = z_a \in U_\beta \cap N_y \subseteq p^{-1}(B). \text{ This is a contradiction since } N_x \cap N_y = \emptyset.$$

Thus, X is Hausdorff.

($\Leftarrow$)

Assume X is an n -manifold and p has countable degree.

Let $\tilde{x} \in \tilde{X}$, thus consider $p(\tilde{x}) = x$. Since p is a covering map and X is an n -manifold. This implies there exists an open set U of x that is evenly covered and open set V of x that is homeomorphic to an open subset of $\mathbb{R}^n$.

Thus, $p^{-1}(U) = \bigcup_{\alpha \in J} V_\alpha$ for disjoint open sets that are homeomorphism to U .

Thus, consider V_α for the $\alpha \in J$ s.t. $\tilde{x} \in V_\alpha$. This implies that $p|_{V_\alpha}: V_\alpha \rightarrow U$ is a homeomorphism.

Since V is open and homeomorphic to an open subset of $\mathbb{R}^n$ this implies that $V \cap U$ is also open and homeomorphic to open subset of $\mathbb{R}^n$. Thus, the open set $V_\alpha \cap p^{-1}|_{V_\alpha}(V)$ is an open neighborhood of $\tilde{x}$ that is homeomorphic to $U \cap V \implies$ it is homeomorphic to an open subset of $\mathbb{R}^n$.

Thus, $\tilde{X}$ is locally homeomorphic to $\mathbb{R}^n$.

Let β be a countable basis for X .

Define $\mathfrak{B} = \{B \in \beta \mid B \text{ is evenly covered}\}$.

Let $x \in X$ then there is an open set s.t. U s.t. U is evenly covered. Thus, there exists a $B \in \beta$ s.t. $x \in B \subseteq U$.

Proven above since B is an open subset of an evenly covered set U this implies that B is evenly covered.

Thus, $\mathfrak{B}$ covers X .

Let $x \in B_1 \cap B_2$ s.t. $B_1, B_2 \in \mathfrak{B}$. Thus, $B_1 \cap B_2$ is evenly covered since it is an open subset of B_1 which is evenly covered. Thus, there exists a $B_3 \in \beta$ s.t. $x \in B_3 \subset B_1 \cap B_2$. Thus, $B_3 \in \mathfrak{B}$ since it is an open subset of an evenly covered set.

Therefore, $\mathfrak{B}$ is a countable basis for X .

Consider $p^{-1}(B)$ for $\mathfrak{B}$ since B is evenly covered $p^{-1}(B) = \sqcup_{\alpha \in J} V_\alpha$ s.t. $V_\alpha \cong B$. Define $V_B = \{V_\alpha \mid \alpha \in J\}$. Now define $\tilde{\mathfrak{B}} = \bigcup_{B \in \mathfrak{B}} V_B$.

Since p has countable degree this implies that V_B is countable thus $\tilde{\mathfrak{B}}$ is countable since it is a countable union of countable sets.

Since $\mathfrak{B}$ covers X and p is surjective this implies that $\tilde{\mathfrak{B}}$ covers $\tilde{X}$.

Let $\tilde{x} \in B_1 \cap B_2$ for $B_1, B_2 \in \tilde{\mathfrak{B}}$. By definition, we know that $B_1 \cong B$ and $B_1 \cong B'$ for some $B, B' \in \mathfrak{B}$. Thus, this implies that $p(\tilde{x}) \in B \cap B' \implies$ there is a $B'' \in \mathfrak{B}$ s.t. $p(\tilde{x}) \in B'' \subset B \cap B'$.

Thus, consider $p|_B^{-1}$ which is a homeomorphism onto B_1 . Consider the open set $p|_B^{-1}(B'') \in \tilde{\mathfrak{B}}$. This is an open set s.t. $\tilde{x} \in p|_B^{-1}(B'') \subset B_1 \cap B_2$.

Thus, $\tilde{\mathfrak{B}}$ is a countable basis for $\tilde{X}$.

Let $x, y \in \tilde{X}$ s.t. $x \neq y$. Consider $p(x), p(y)$:

If $p(x) \neq p(y)$ then let A, B be two disjoint open sets s.t. $p(x) \in A$ and $p(y) \in B$. Then let U and V be the evenly covered neighborhoods of $p(x)$ and $p(y)$.

Consider $p^{-1}(A \cap U)$ and $p^{-1}(B \cap V)$. Since $p^{-1}(A \cap U) \cap p^{-1}(B \cap V) = p^{-1}(A \cap U \cap B \cap V) = \emptyset$. This implies that $p^{-1}(A \cap U)$ and $p^{-1}(B \cap V)$ are disjoint open neighborhoods of x, y .

If $p(x) = p(y)$ then let U be the evenly covered neighborhood of $p(x)$.

Thus, $p^{-1}(U) = \sqcup_{\alpha \in J} V_\alpha$ s.t. $V_\alpha \cong U$. Since $V_\alpha \cong U$ and $x \neq y$. This implies that $x \in V_\alpha$ and $y \in V_\beta$ for $\alpha \neq \beta$. Since if $\alpha = \beta$ then this implies that $p^{-1}|_{V_\alpha}(p(x)) = x = y$ which is a contradiction.

Thus, $\tilde{X}$ is an n -manifold. ■

Problem 5:

Fix a basepoint $x_0 \in S^1$, and let $\phi : \pi_1(S^1, x_0) \rightarrow \pi_1(S^1, x_0)$ be any group homomorphism. Prove that there is a continuous map $f : S^1 \rightarrow S^1$ so that $\phi = f_*$.

Proof.

Since S^1 is path connected we know that $\pi_1(S^1, x_0) \cong \pi_1(S^1, (1,0))$. Thus, make our basepoint $(1,0)$

Proven in class $\pi_1(S^1, (1,0)) \cong \mathbb{Z}$. This, implies that $\pi_1(S^1, (1,0))$ is cyclic and generated by a single element. Thus, let $\pi_1(S^1, (1,0)) = \langle [\alpha] \rangle$.

Since ϕ is a homomorphism this map will be defined by where it sends $[\alpha]$. Thus, $\phi([\alpha]) = [\beta]$, but since $[\alpha]$ generates $\pi_1(S^1, (1,0))$. This implies that $[\beta] = [\alpha]^k = ([\alpha] * [\alpha] * [\alpha] * \dots * [\alpha])$.

Thus, $\phi([\alpha]) = [\alpha]^k$ for some $k \in \mathbb{Z}$.

Define $f_k : S^1 \rightarrow S^1$ s.t. $f_k(\theta) = (\cos(k\theta), \sin(k\theta))$. This is continuous since $\cos, \sin$ are continuous.

Thus, $(f_k)_*([\alpha]) = [f_k \circ \alpha] = [\alpha]^k = \phi([\alpha])$.

Since proven in class $(f_k)_*$ is a homomorphism, and it is defined where it sends $[\alpha]$.

This implies that $(f_k)_* = \phi$. ■